P **Plant Ride**

SRM ECO TECH · FOR LHS

MVP scope

What the Plant Ride MVP delivers, screen by screen: the **user** (employee) app first, then the **driver** app, then the **admin** console at the transport desk. Every screen is shown as it runs today, with what it is for, what is in scope, the rules it enforces and what is left for a later phase.

Part 1 · 12 screens

Users

Employees book, board with an OTP, ride, rate and see the charge to their cost centre.

Part 2 · 8 screens

Driver

Start-of-duty compliance check, pickups in route order, OTP boarding, issues, sign-off.

Part 3 · 7 screens

Admin

Live fleet board, queue and approvals, billing by cost centre, routes, rules and audit.

Product	Plant Ride: smart cab booking, pooling, fleet compliance and cost-centre billing inside the LHS plant
Scope baseline	Proposal decks *Plant Ride: Solution Proposal* and *User Journey Walkthrough*; MVP = the deck's Phase 2 pilot scope plus cost-centre billing
Status	Built and verified; screenshots taken from the running MVP
Date	30 September 2026

OVERVIEW

Scope at a glance

The MVP covers one site, the three configured routes and the full trip lifecycle: booking, matching, pooling within the occupancy cap, OTP boarding, drop, cost allocation and audit. It runs as one Expo app (Android, iOS and web) on a Spring Boot API with MySQL.

3
user roles with their own app: user, driver, admin (plus a desk-only role)

27
screens and screen states in scope, shown in this document

3
routes, 9 vehicles, own and vendor fleet in the demo data

3
riders per pick-up run at most (configurable per route)

Who uses what

ROLE	SIGNS IN WITH	DEVICE	SCREENS
User (employee rider)	P. No. and password	Phone (Android, iOS) or browser	Home, Book, Ride options, Booking, My trips, Alerts, Me
Driver	Driver ID and password	Phone, large touch targets	Trips (duty check, queue, boarding), Issues, Duty
Admin (fleet admin and desk)	Username and password	Web console on a laptop or desk monitor	Live board, Queue, Billing, Routes & rules
Desk operator	Username and password	Web console or gate tablet	Live board, Queue (no billing or configuration)

Contents

Part 1 Users (employee app): U1–U12 page 4

Part 2 Driver app: D1–D8 page 10

Part 3 Admin console: A1–A7 page 15

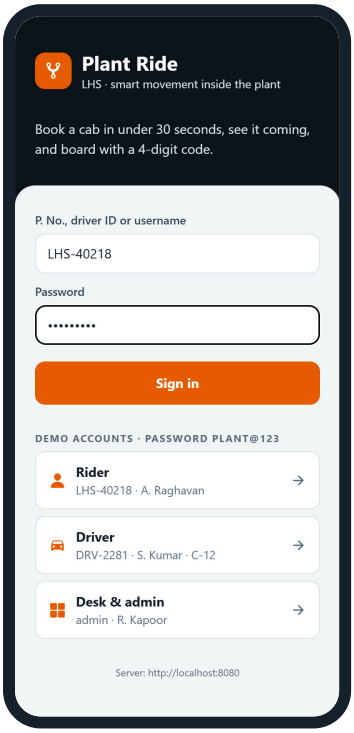
Part 4 Business rules, not in the MVP, and how the scope was verified page 22

Scope against the proposal

CAPABILITY IN THE PROPOSAL	MVP	HOW IT IS DELIVERED
Book in under 30 s with HR-prefilled, editable details; book for a guest	**In scope**	Employee record stands in for HRMS; guest rides charge the host's cost centre
Ride options: shuttle (information), shared, exclusive with approval	**In scope**	Shared riders join cabs already on the route (pooling)
Scheduled rides up to 7 days ahead	**In scope**	Dispatched 20 min before pickup; the app offers five quick time choices
Route sequencing, stops in order, leg times, geofences	**In scope**	Master data, editable by the admin without a release
Occupancy rule: max 3 per pick-up run	**In scope**	Hard rule per route segment; cannot be overridden by the desk
Matching engine that explains its decisions	**In scope**	Filter, score, suggest; the reason is stored on every booking
Confirmation with vehicle number, driver contact, OTP, tracking link	**In scope**	Tracking link shows the vehicle only and expires with the trip
Driver app: sign-on check, route-order queue, OTP, no-show, issues, SOS	**In scope**	GPS pings while on duty; geofence arrival at stops
Fleet compliance with expiry alerts and a block on expiry	**In scope**	Alerts at 30, 7 and 1 days; expired documents block duty and dispatch
Command centre: fleet on one screen, KPIs, alerts, overrides	**In scope**	Web console, refreshed every 5 s
Cost allocation by cost centre, budgets, ERP export, vendor bill vs GPS	**In scope**	Distance-based split; CSV export shaped for SAP
Notifications and audit trail	**In scope**	In-app notifications; every state change audited
Hindi UI, offline driver app, SSO/HRMS/SAP live integrations, plant map, demand forecasting	**Later phase**	See the last section, *Not in the MVP*

Users (employee) screens

The employee's app: book in under 30 seconds, see the cab and driver, share a tracking link, board with a 4-digit OTP, rate the ride and see what it cost. Four tabs: **Home**, **Rides**, **Alerts**, **Me**.



U1 Sign in

One sign-in screen for every role; the app opens the right side for the person.

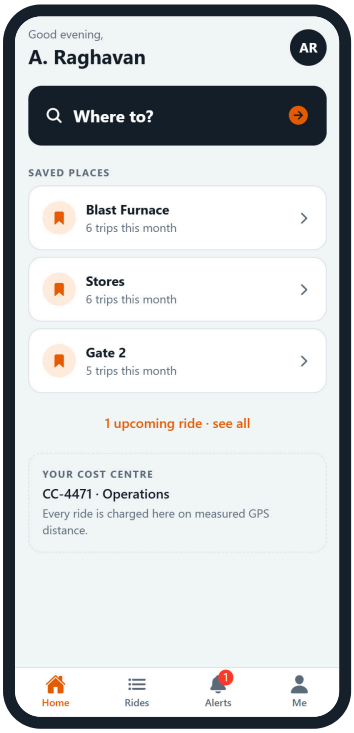
IN SCOPE

- Sign in with P. No. (users), driver ID (drivers) or username (desk)
- One-tap demo accounts below the form
- Session lasts 12 hours (one shift); server address shown for support

RULES

- Each role lands in its own app and cannot open another role's screens.

Not in the MVP: SSO or Active Directory login; password reset flow.



U2 Home

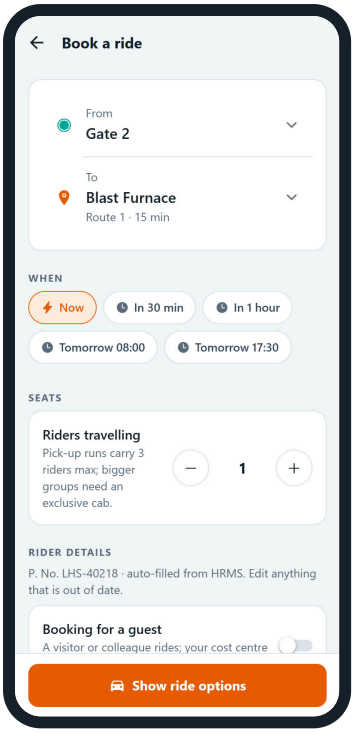
The starting point for every ride.

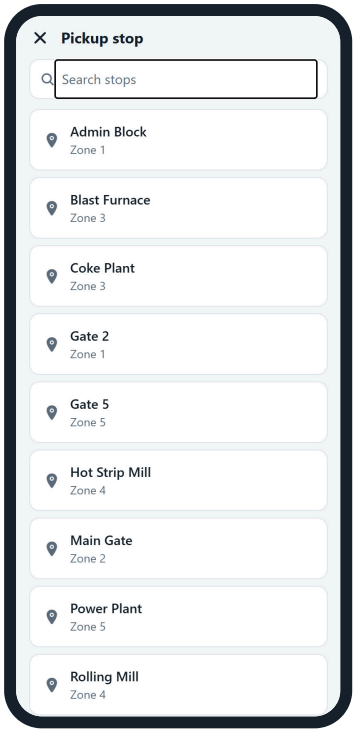
IN SCOPE

- **Where to?** button opens booking
- Saved places with this month's trip counts; one tap starts a booking
- Upcoming ride shortcut
- The employee's cost centre, shown before booking

RULES

- Every ride is charged to this cost centre on measured GPS distance.
- Booking is disabled while the employee is paused for no-shows (U12).





U3 Book a ride

Choose where from, where to, when and how many riders on one screen.

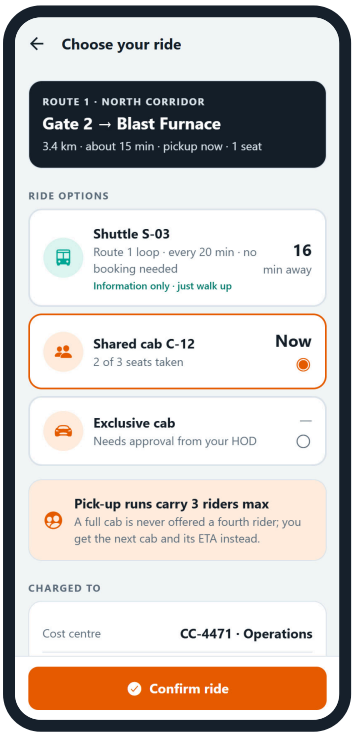
IN SCOPE

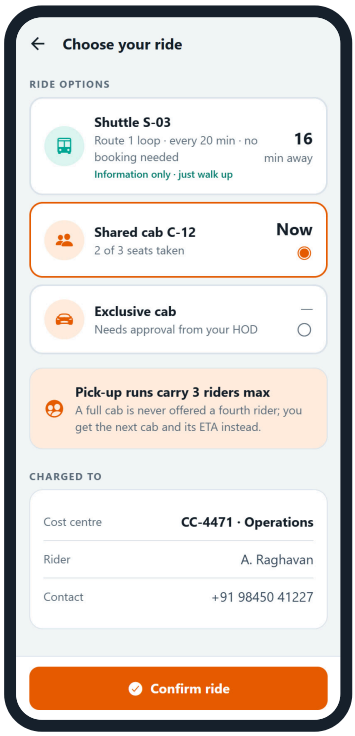
- From and To from a searchable stop list with zones
- When: Now, In 30 min, In 1 hour, Tomorrow 08:00, Tomorrow 17:30
- Seats (riders travelling)
- Rider details prefilled from HR, editable; **Booking for a guest**

RULES

- The route is resolved from the stop sequence.
- Pick-up runs carry 3 riders at most; a larger group needs an exclusive cab.
- A guest ride is charged to the host's cost centre.

Not in the MVP: a free date and time picker in the app (the API accepts any time up to 7 days ahead); recurring bookings.





U4 Choose your ride

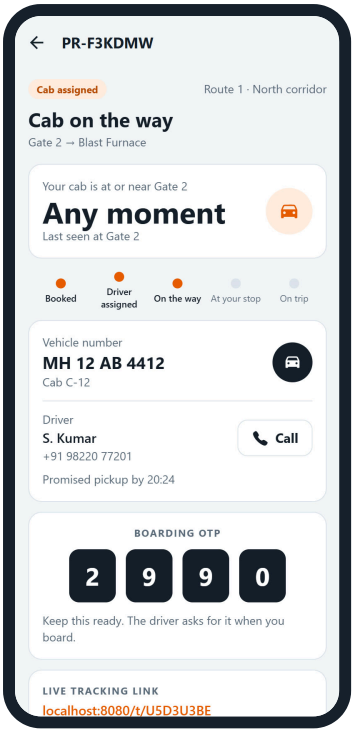
Compare the ways to travel and confirm one.

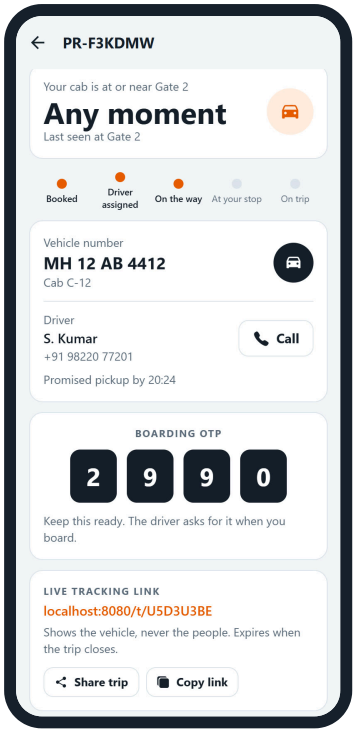
IN SCOPE

- Route summary: distance, time, pickup, seats
- **Shuttle**: timetable information only, walk up
- **Shared cab**: the cab, its ETA and seats taken
- **Exclusive cab**: whole cab, needs approval
- Cost centre, rider and contact shown before **Confirm ride**

RULES

- A full cab is never offered; the rider gets the next cab and its ETA.
- A repeated tap on Confirm returns the same booking, never a duplicate.





U5 Ride confirmed

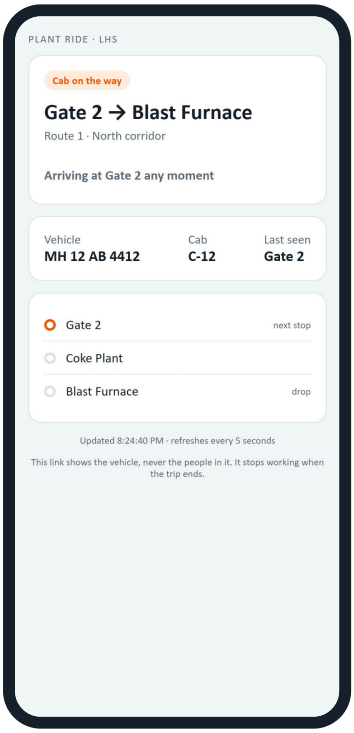
Everything the rider needs to meet the cab, without phoning the desk.

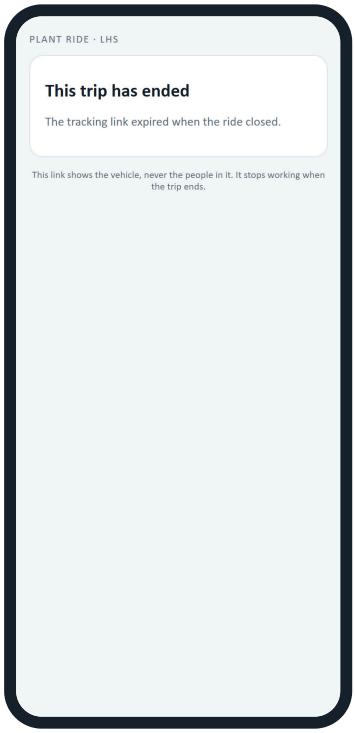
IN SCOPE

- Booking code and live status with a five-stage progress bar
- Vehicle number, cab code, driver name and **Call**
- Promised pickup time
- 4-digit **boarding OTP**
- **Live tracking link** with Share trip and Copy link
- **Cancel ride** before boarding

RULES

- The screen refreshes every 5 seconds.
- An exclusive request shows *Waiting for approval* until the desk or HOD releases it.





U6 Shared tracking page

A public page anyone with the link can open, with no login.

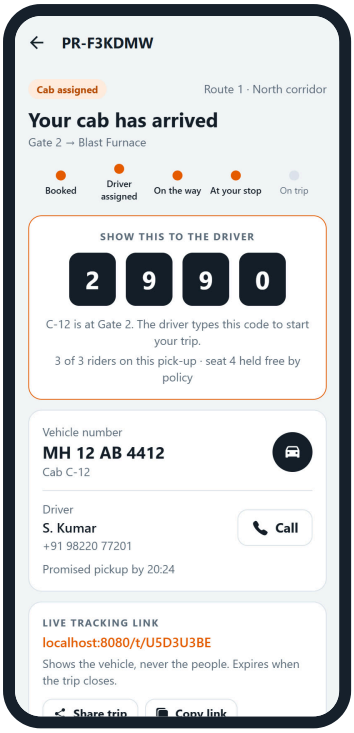
IN SCOPE

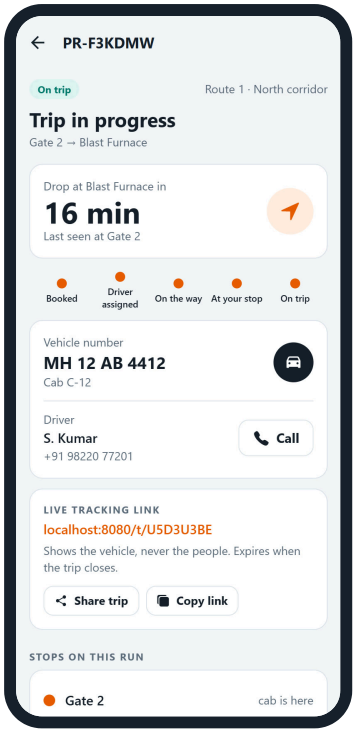
- Stage, route, vehicle number, cab code, last seen stop, stop list
- Refreshes every 5 seconds
- *This trip has ended* once the ride closes

RULES

- Tracks vehicles, not people: no names or phone numbers are served.
- The link expires when the trip closes.
- The page cannot be embedded in another site.

Not in the MVP: a live map of the plant roads (stop progress is schematic).





U7 Cab arrived and trip in progress

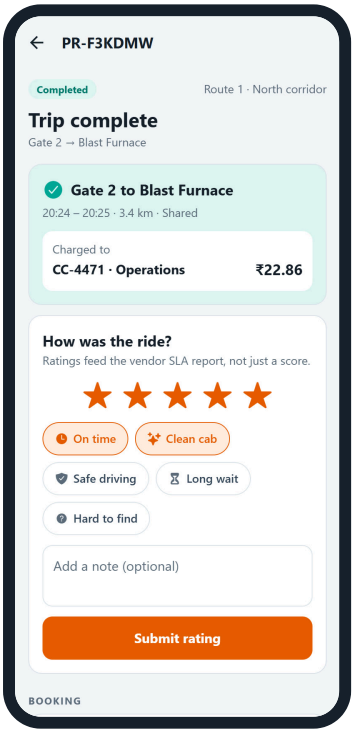
The booking screen changes by itself as the trip moves on.

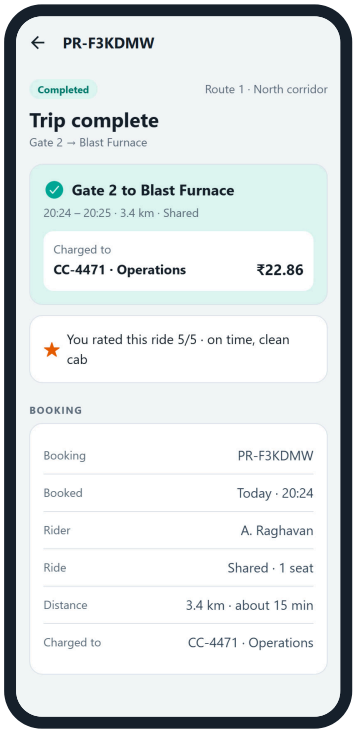
IN SCOPE

- **Your cab has arrived**: OTP in large digits to show the driver, riders on this pick-up
- **Trip in progress**: drop ETA, vehicle, driver, stops on this run, co-riders

RULES

- The cab waits the route's wait time (5 min on Route 1); then the driver may mark a no-show.
- Seats above the cap are held free by policy.





U8 Trip complete and rating

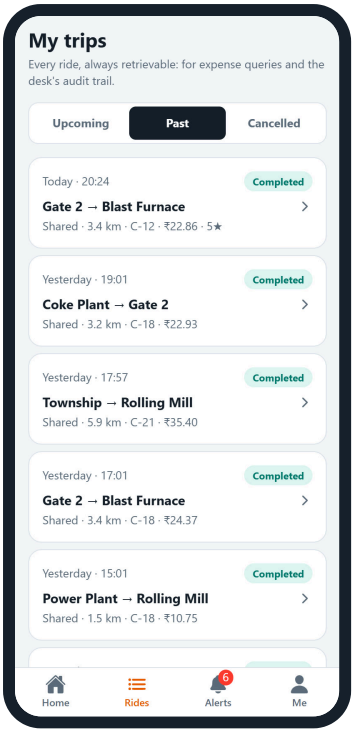
Close the ride: what it cost, who paid, and how it went.

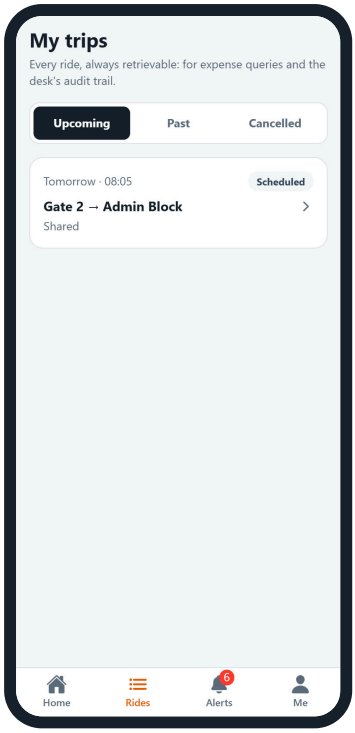
IN SCOPE

- Route, times, distance and ride type
- Amount charged and the cost centre
- 1–5 stars, quick tags (On time, Clean cab, Safe driving, Long wait, Hard to find) and a note
- Booking record: code, rider, distance, cost centre

RULES

- Trip cost = GPS route km × rate card, split across riders by distance ridden; shares add up to the paisa.
- Ratings feed the vendor SLA report.





U9 My trips

Every ride the employee has taken or booked, for expense queries.

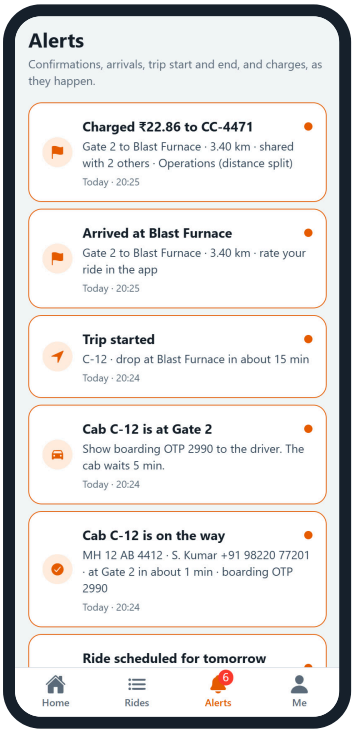
IN SCOPE

- Upcoming, Past and Cancelled tabs
- Each ride: date, route, distance, cab, cost, rating; tap for the full record

RULES

- Ride history is kept for billing and safety only.

Not in the MVP: downloadable monthly statement for the employee.



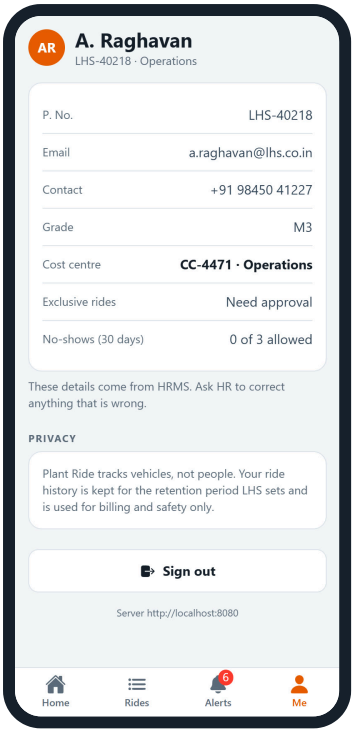
U10 Alerts

Every step of a ride as a notification.

IN SCOPE

- Cab assigned (vehicle, driver, phone, OTP), cab at the stop, trip started, arrived, charged
- Scheduled ride confirmations; unread badge on the tab
- Delivery status logged for every notification

Not in the MVP: push, SMS and WhatsApp delivery (in-app only; gateways plug in behind the notification service).



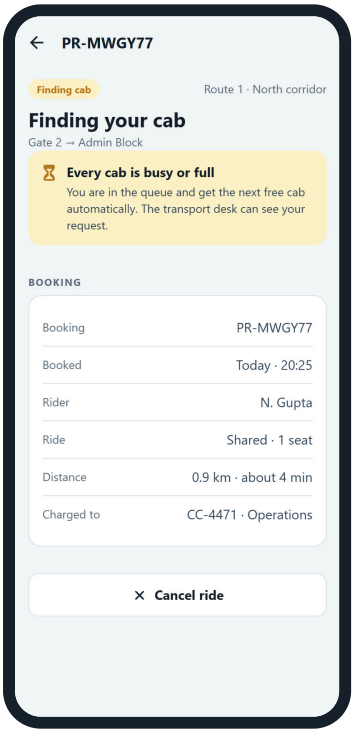
U11 Me

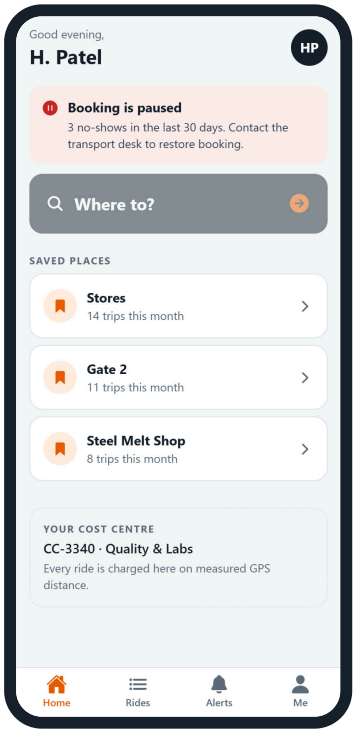
The employee's profile and the rules that apply to them.

IN SCOPE

- P. No., email, contact, grade, cost centre
- Whether exclusive rides need approval
- No-shows in the last 30 days against the allowance
- Privacy note and **Sign out**

Not in the MVP: editing the profile in the app (corrections go through HR); Hindi language option.





U12 Exception states

What the user sees when a ride cannot go ahead normally.

IN SCOPE

- **Every cab is busy or full**: the rider waits in the queue and gets the next free cab automatically; the desk sees the request
- **Booking is paused**: after 3 no-shows in 30 days; the rider contacts the desk
- **Waiting for approval**: exclusive ride until released
- Desk cancellation shows the desk's reason

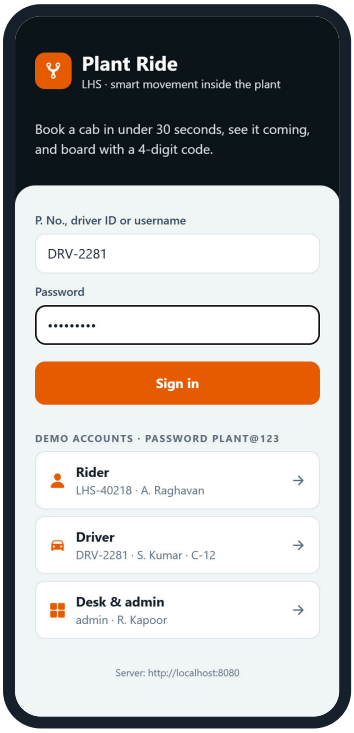
RULES

- The engine retries waiting requests every 15 seconds; unserved requests expire after 90 minutes.

PART 2 · DRIVER APP

Driver screens

Built for use in the cab: large targets usable with gloves, pickups in route order, and a permanent SOS button while on duty. Three tabs: **Trips**, **Issues**, **Duty**.

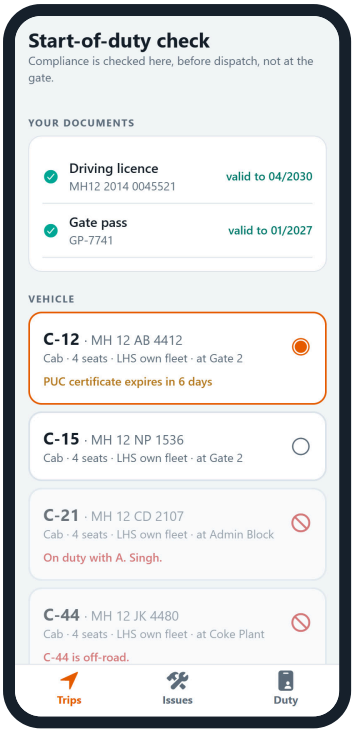


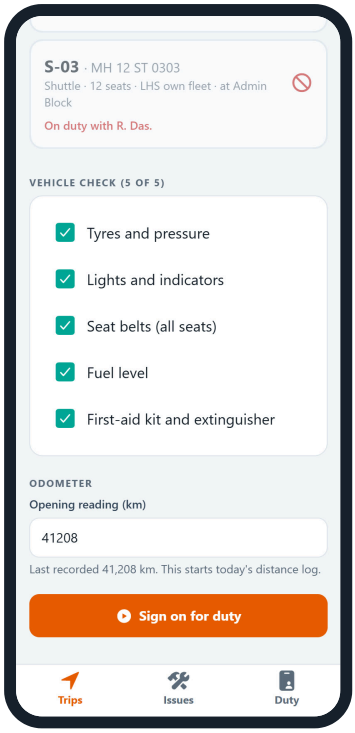
D1 Sign in

The shared sign-in screen, with the driver ID.

IN SCOPE

- Driver ID and password; 12-hour session
- Lands on the start-of-duty check, or on the trip queue if already on duty





D2 Start-of-duty check

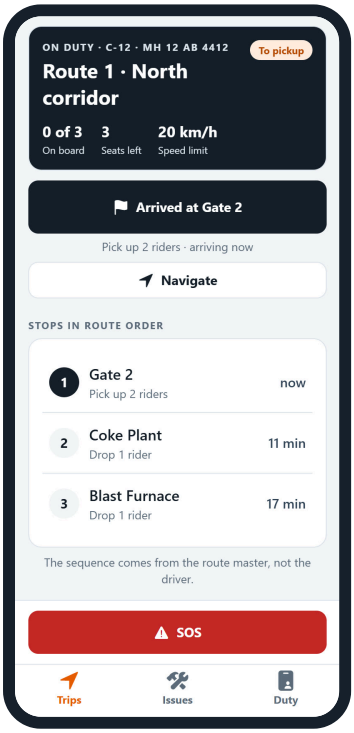
Compliance is checked here, before dispatch, not at the gate.

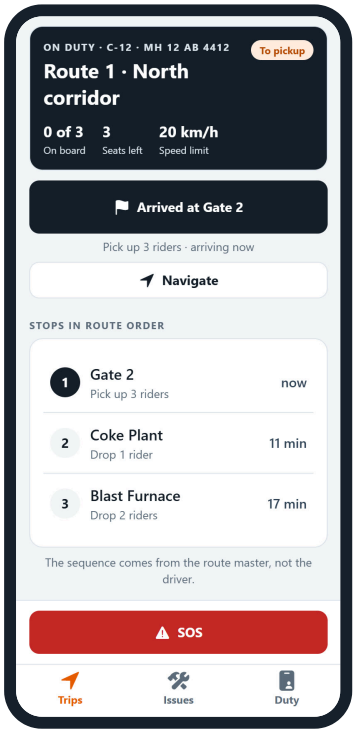
IN SCOPE

- Driver's documents with validity: driving licence, gate pass
- Vehicle choice; unavailable vehicles greyed out with the reason
- Vehicle documents flagged ahead of expiry (e.g. PUC in 6 days)
- Five-point check: tyres, lights, seat belts, fuel, first-aid kit and extinguisher
- Opening odometer, prefilled from the last reading

RULES

- An incomplete check is refused.
- Expired licence, gate pass, insurance, fitness or PUC blocks duty (D8).
- One driver per vehicle and one vehicle per driver at a time.





D3 Trip queue

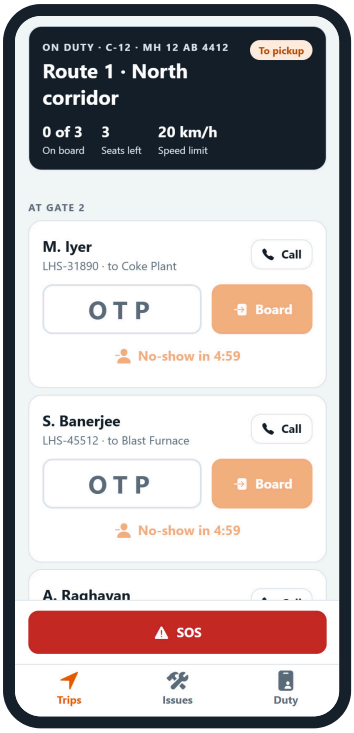
The screen a driver lives in all day.

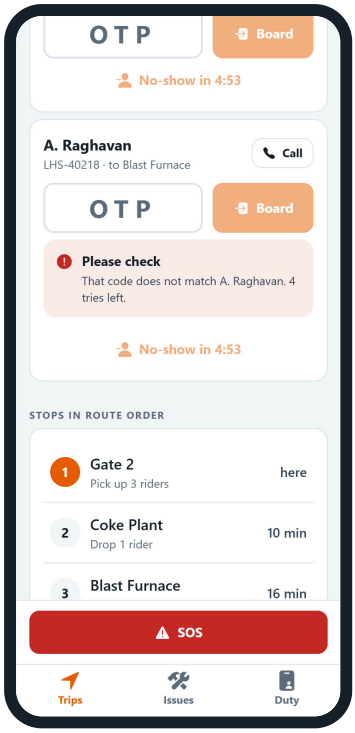
IN SCOPE

- Header: vehicle, route, on board, seats left, speed limit, status
- Next action as one large button, e.g. **Arrived at Gate 2**
- Stops in route order with pickups, drops and ETAs
- **Navigate** (opens map directions) and **SOS**
- New riders appear automatically
- *Slow down* banner on overspeed; *Location is off* banner

RULES

- Riders already waiting are matched the moment the driver signs on.
- The sequence comes from the route master, not the driver.
- GPS pings are sent while on duty; overspeed alerts the desk.





D4 At the stop: OTP boarding and no-show

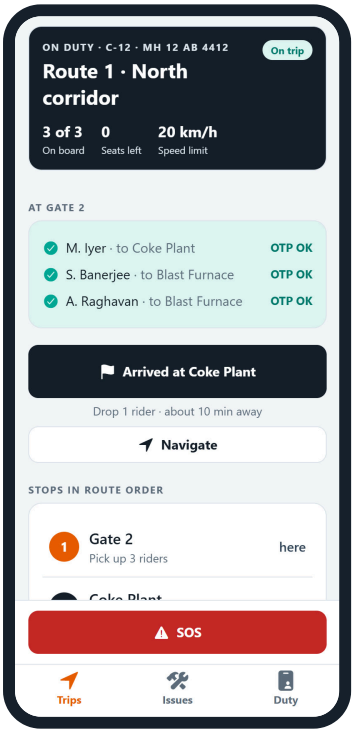
Board the right rider into the right cab.

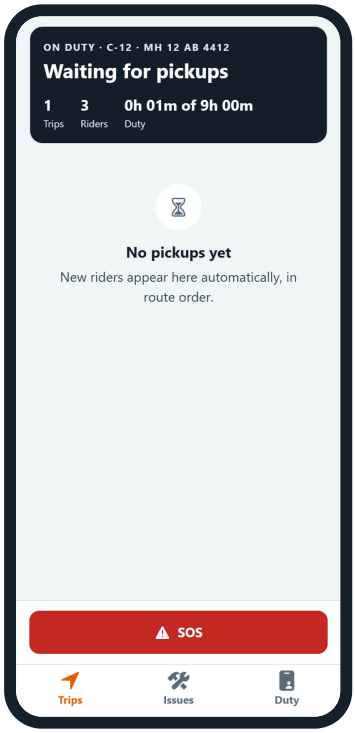
IN SCOPE

- One card per rider: name, P. No., destination, **Call**
- OTP entry and **Board**
- No-show countdown, then **Mark no-show**
- Clear message on a wrong code with tries left; *Boarding locked* after too many

RULES

- 4-digit OTP per booking; 5 wrong codes lock boarding for that rider.
- The driver cannot leave a stop while a rider there is still waiting.
- No-show allowed only after the route's wait time.





D5 On trip, drops and trip close

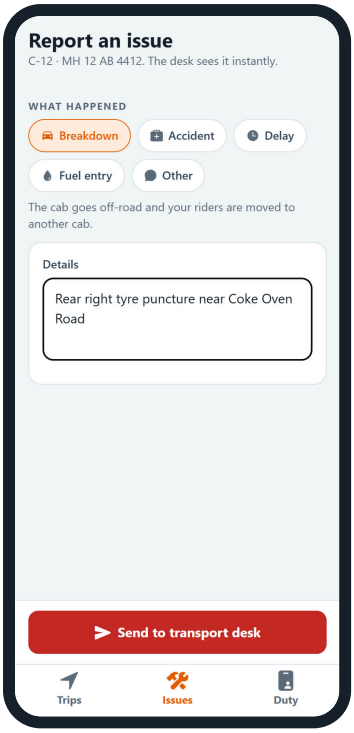
Drive the run and drop each rider in order.

IN SCOPE

- Boarded riders listed as *OTP OK*; on board count updates
- **Arrived at ...** for each drop, with distance and time to it
- After the last drop: *Waiting for pickups* with today's trips, riders and duty time

RULES

- A GPS ping inside the stop's geofence records arrival automatically; the button works everywhere.
- The last drop closes the trip, calculates the cost and expires tracking links.



D6 Issues and SOS

Tell the desk what went wrong, from the cab.

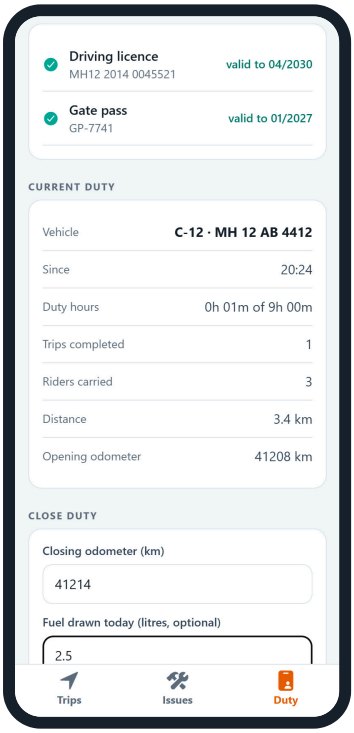
IN SCOPE

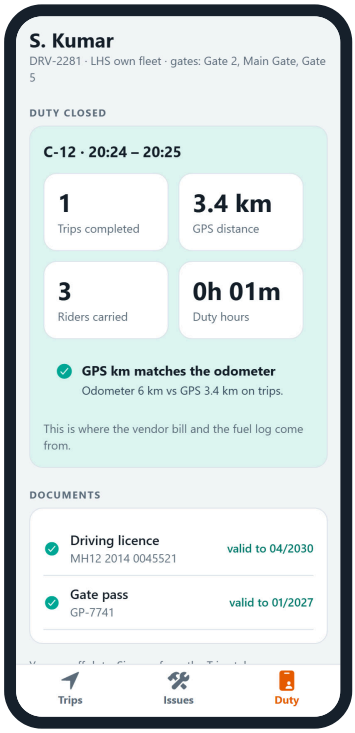
- Breakdown, Accident, Delay, Fuel entry, Other, with details
- **Send to transport desk**
- **SOS** (Trips tab) sends cab and location to the desk and plant security, after a confirmation

RULES

- A breakdown takes the cab off-road and puts its riders back in the queue from the last stop reached.
- Every issue raises a desk alert.

Not in the MVP: photo attachments; offline buffering of issues and GPS.





D7 Duty and sign-off

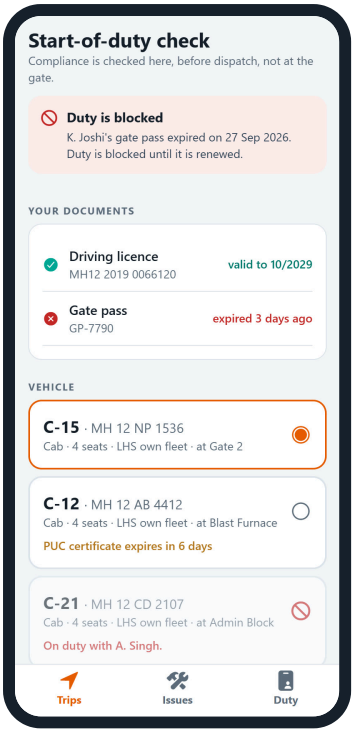
End the shift with a distance check the vendor bill relies on.

IN SCOPE

- Current duty: vehicle, start time, duty hours against the limit, trips, riders, distance, opening odometer
- Close duty: closing odometer and fuel drawn
- Duty closed summary with the odometer-vs-GPS check

RULES

- *Finish your trip first*: sign-off is blocked with riders on board.
- Duty limit 9 hours; the engine will not assign a trip that would break it.
- GPS km is the contractual source for billing.



D8 Duty blocked

An expired document stops the driver before they reach a rider.

IN SCOPE

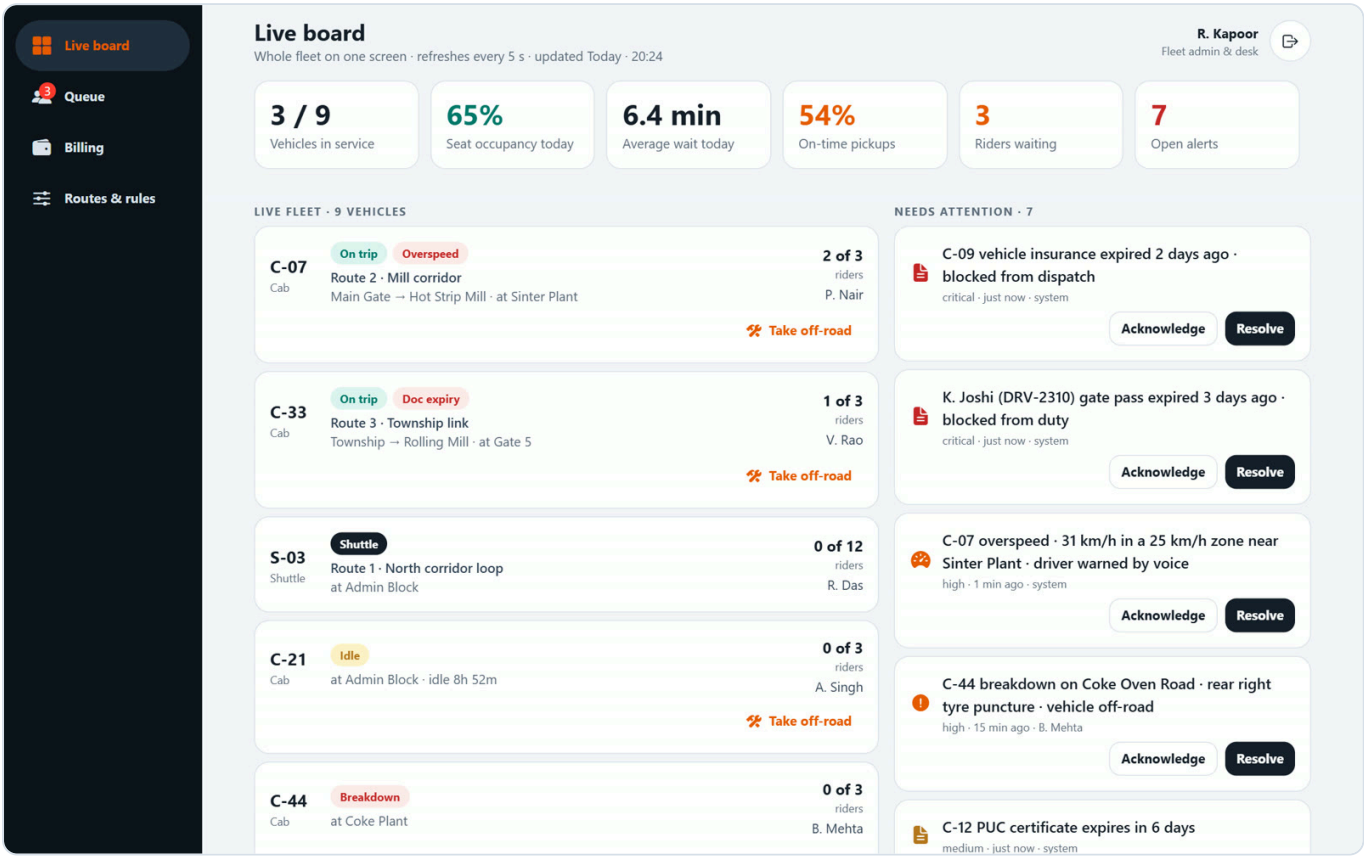
- *Duty is blocked* with the document and its expiry date
- Expired items shown in red on the document list

RULES

- Expiry blocks duty and dispatch; it is not just a warning.
- Alerts go to the desk at 30, 7 and 1 days before expiry and on expiry.

Admin screens

The transport desk's console, in a browser on a laptop or desk monitor (a tablet layout works at the gate). The admin sees four sections; a desk operator sees the first two.



A1 Live board

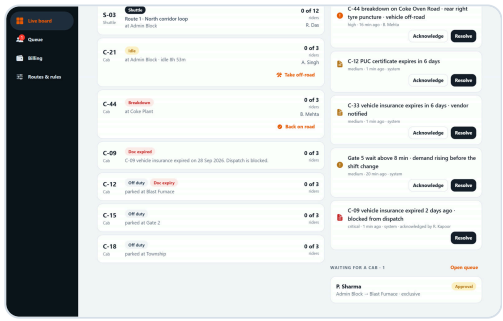
The whole fleet on one screen, refreshed every 5 seconds.

IN SCOPE

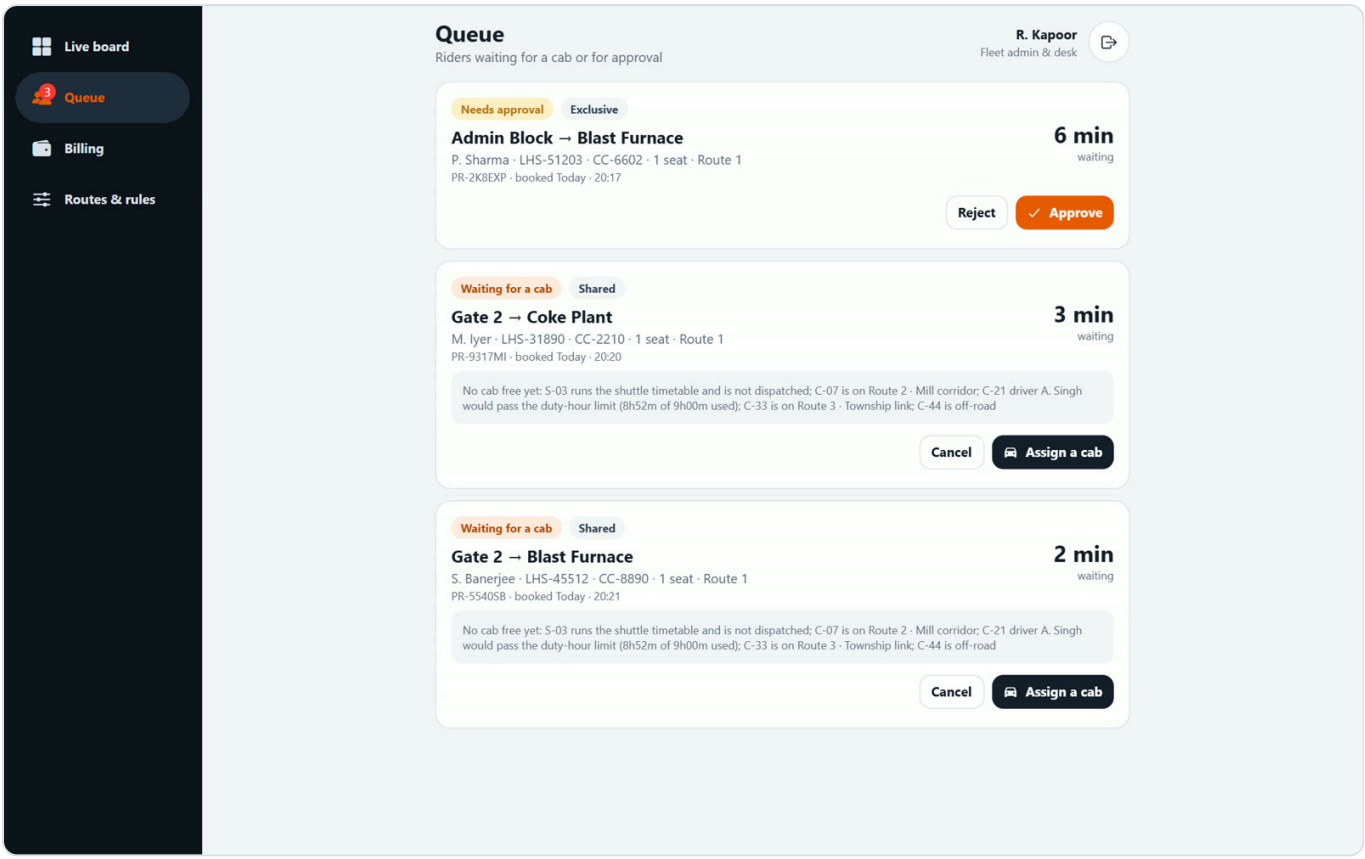
- KPIs: vehicles in service, seat occupancy, average wait, on-time pickups, riders waiting, open alerts
- Every vehicle: status, route, position, riders on board, driver
- **Take off-road** / **Back on road**
- Needs attention: document expiry, overspeed, breakdown, SOS, long waits; **Acknowledge** and **Resolve**
- Waiting riders with a link to the queue

RULES

- Acknowledging an alert does not unblock a vehicle with an expired document.
- Each alert shows its severity, age and source.



Alert acknowledged by the admin



A2 Queue and approvals

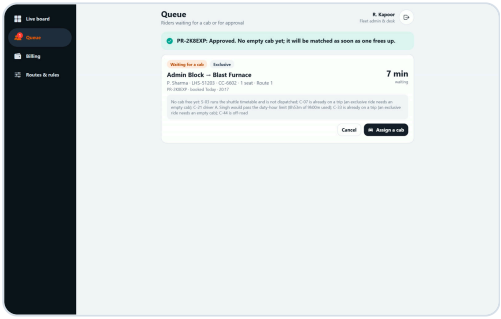
Every rider waiting for a cab or for approval, with the engine's reason.

IN SCOPE

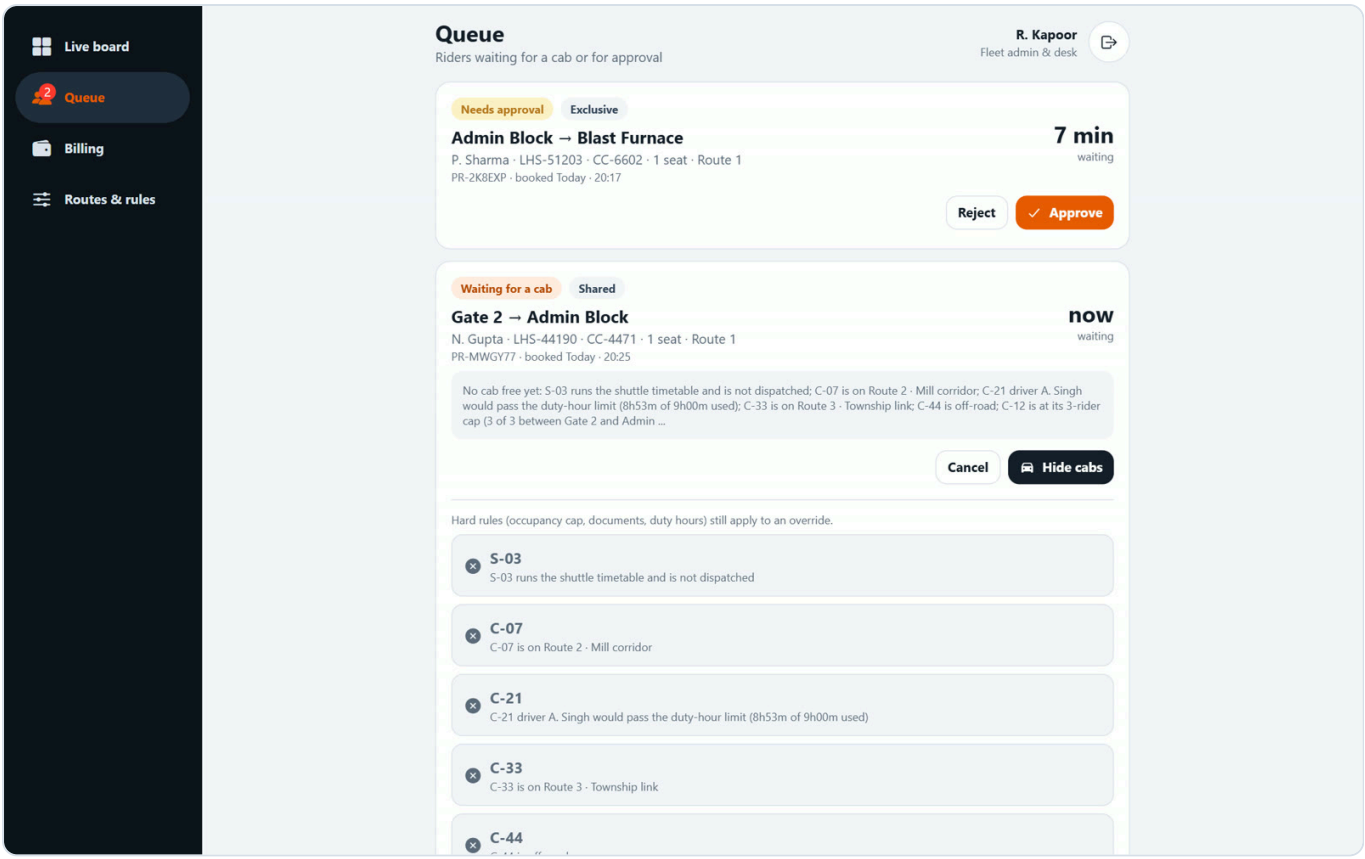
- Request cards: status, ride type, route, rider, P. No., cost centre, minutes waiting
- The engine's note on why no cab was assigned
- Exclusive rides: **Approve** or **Reject** with a reason
- **Cancel** with a reason shown to the rider

RULES

- An approved exclusive ride waits for an empty cab.
- Every approval, rejection and cancellation is audited.



After Approve: waiting for an empty cab



A3 Assign a cab (desk override)

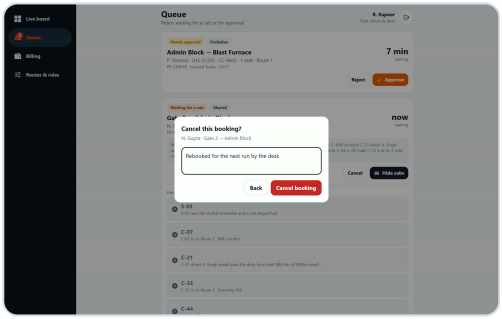
The desk can place a rider on a specific cab, within the hard rules.

IN SCOPE

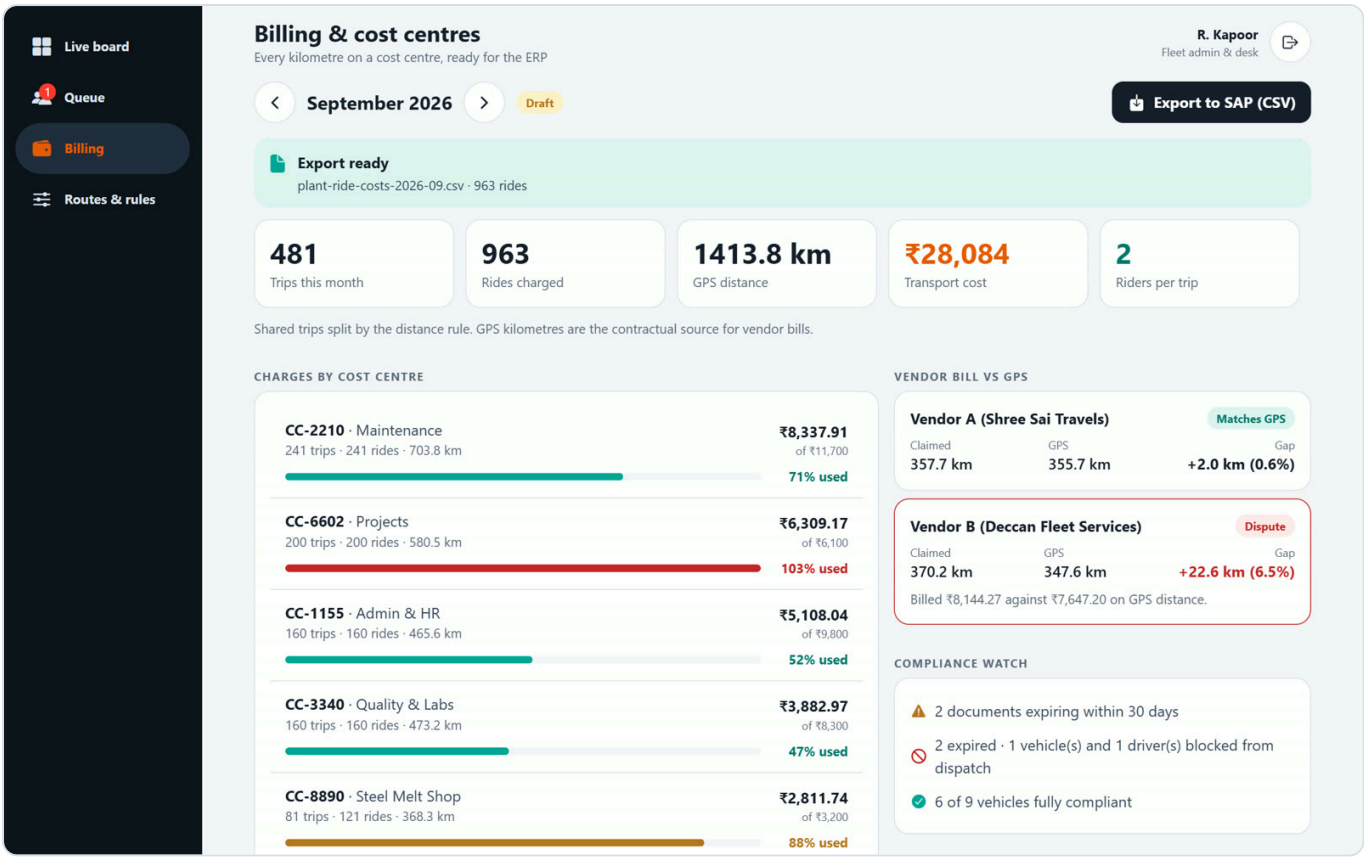
- Every vehicle listed as feasible (ETA, seats taken, **Assign**) or refused with the reason
- Cancel dialog with a reason field

RULES

- Hard rules still apply to an override: occupancy cap, documents, duty hours.
- A cab at its 3-rider cap is refused, e.g. *C-12 is at its 3-rider cap*.



Cancel with a reason shown to the rider



A4 Billing and cost centres

Every kilometre on a cost centre, ready for the ERP. Admin only.

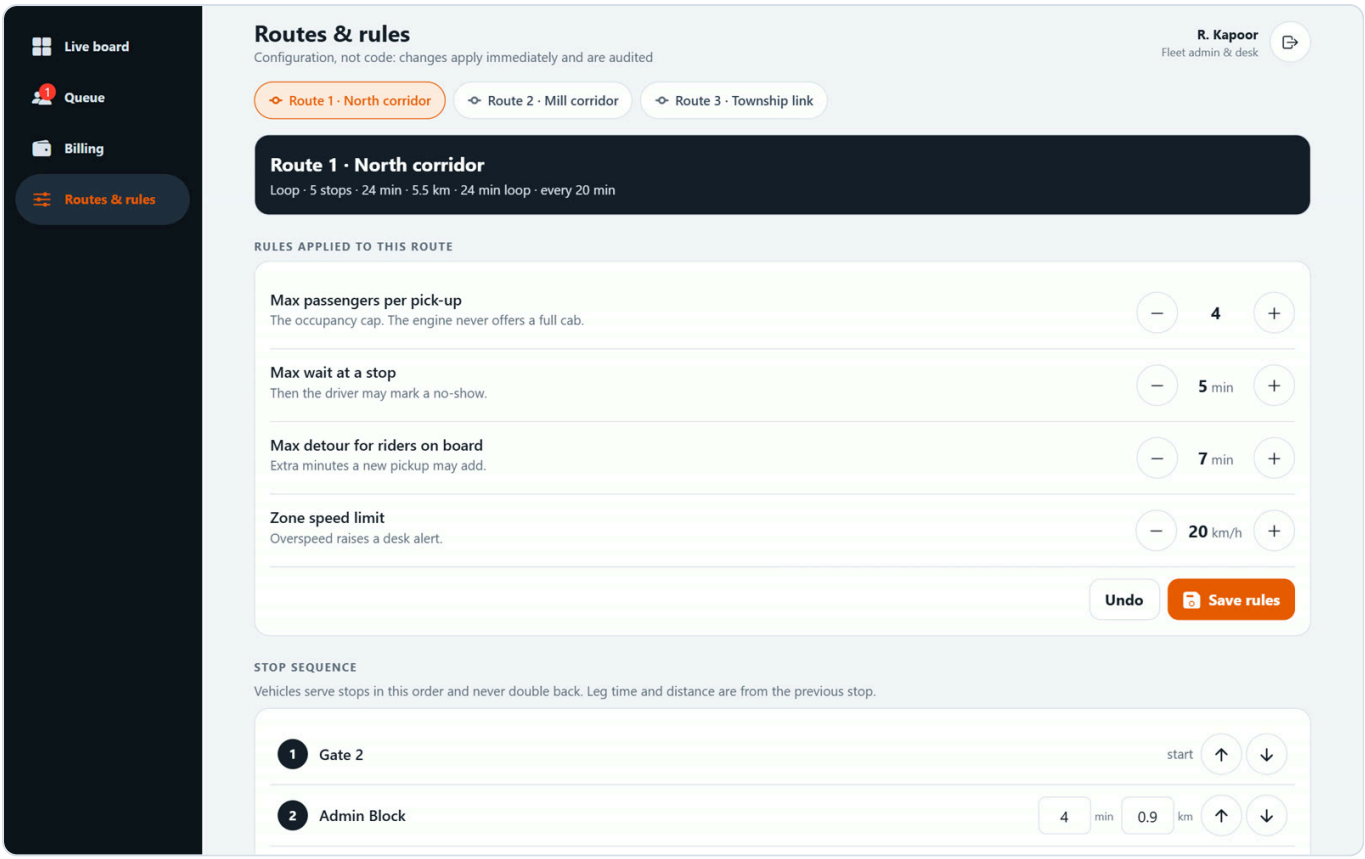
IN SCOPE

- Month selector with Draft or Closed status
- Totals: trips, rides charged, GPS km, cost, riders per trip
- Charges by cost centre with budget bars
- Vendor bill vs GPS, with gap and dispute flag
- Compliance watch; **Export to SAP (CSV)**

RULES

- Budget warning at 80%, over budget at 100%.
- GPS km is the contractual source; a vendor claim above GPS is flagged.
- One cost allocation per ride; no double billing.

Not in the MVP: direct posting to SAP (CSV export only); closing a month from the screen.



A5 Routes and rules

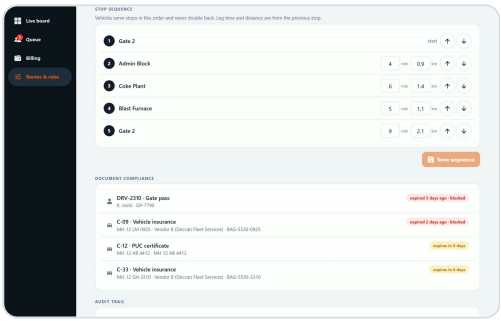
Configuration, not code: changes apply immediately and are audited. Admin only.

IN SCOPE

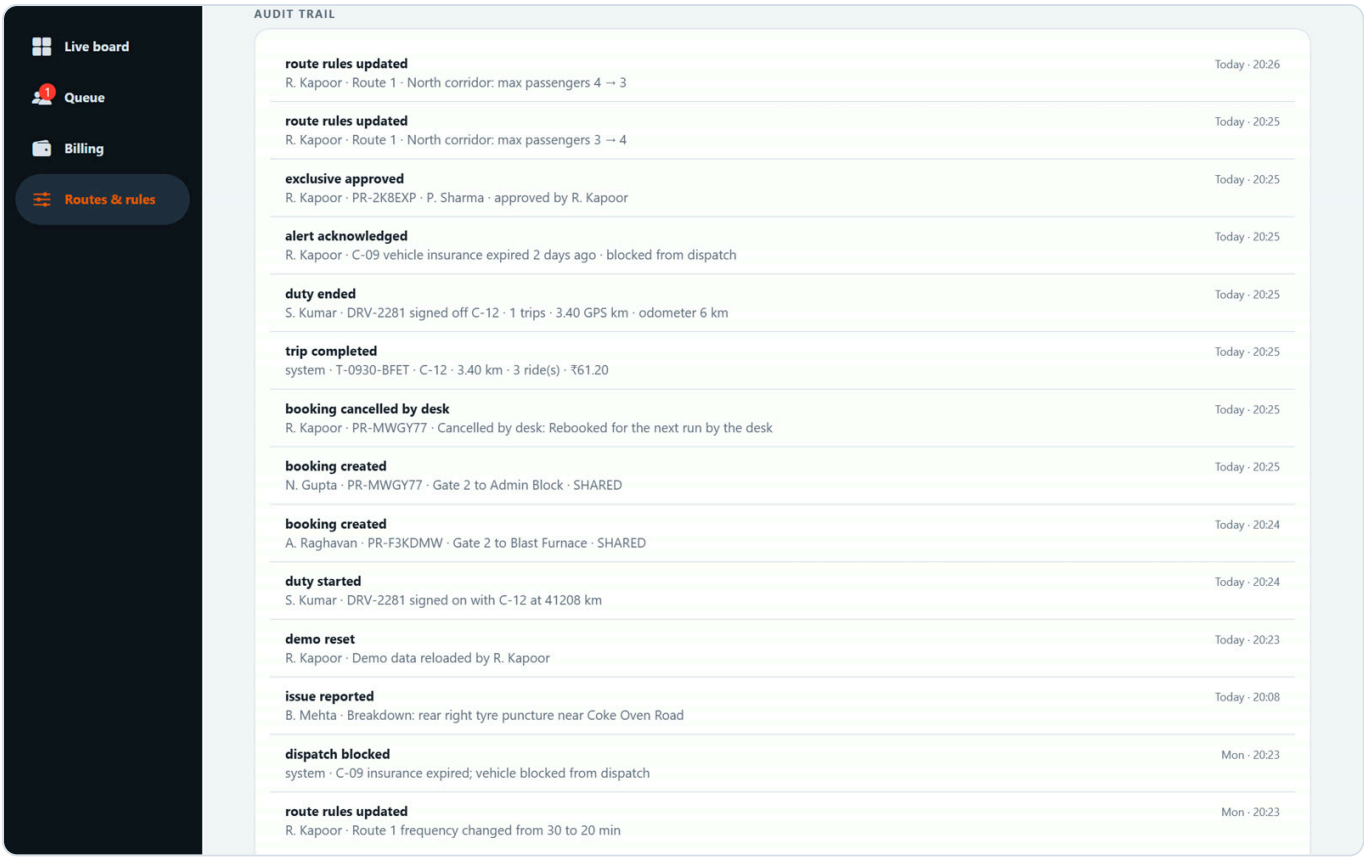
- Route chips; route summary (type, stops, time, distance, frequency)
- Rules per route: max passengers, max wait, max detour, zone speed limit; **Undo** and **Save rules**
- Stop sequence with leg minutes and km; reorder and **Save sequence**
- Document compliance list for drivers and vehicles

RULES

- Saving rules asks for confirmation and shows each change, before and after.
- Two admins cannot overwrite each other: a stale save is refused.
- New trips follow a new sequence; vehicles never double back.



Stop sequence and document compliance



A6 Audit trail and demo tools

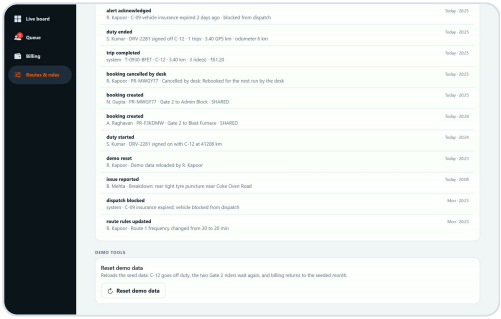
Who did what, when, for every state change.

IN SCOPE

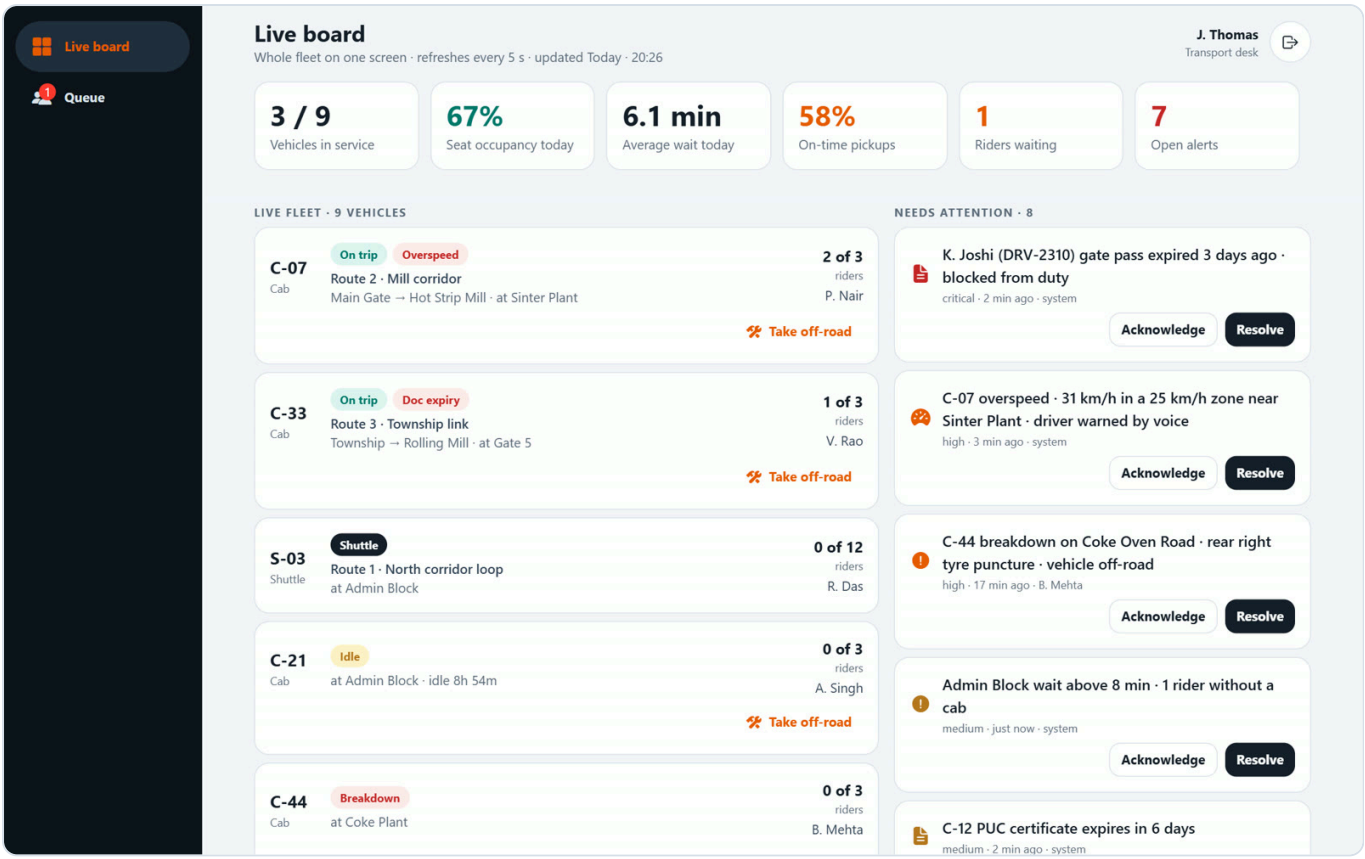
- Bookings, cancellations, approvals, duty start and end, trips, alerts, rule changes, with person and time
- **Reset demo data** (demo builds only)

RULES

- The audit trail is written by the server, not the screens.
- The reset tool is switched off outside the demo.



Demo tools: reset demo data



A7 Roles: admin and desk operator

The same console, trimmed to the person's role.

IN SCOPE

- **Admin**: Live board, Queue, Billing, Routes & rules
- **Desk operator**: Live board and Queue

RULES

- Enforced in the app and on the server.

PART 4

Business rules, not in the MVP, and verification

Business rules in the MVP

RULE	VALUE IN THE MVP	WHERE IT IS SET
Max riders per pick-up run	3	Per route (Routes & rules)
Wait at a stop before a no-show	5 min	Per route
Extra minutes a new pickup may add for riders on board	7 min	Per route
Zone speed limit (overspeed raises a desk alert)	20 km/h	Per route (Route 1)
Wrong OTP attempts before boarding locks	5	System
No-shows that pause booking	3 in 30 days	Setting
Driver duty limit	9 h	Setting
Matching sweep for waiting riders	every 15 s	System
Waiting request expires if unserved	90 min	Setting
Scheduled ride dispatched before pickup	20 min	Setting
Document expiry alerts	30 / 7 / 1 days	System; expiry blocks duty and dispatch
Budget flags	80% / 100%	System
Cost split for shared rides	by distance	Setting (equal split also supported)
Session length	12 h	System

Not in the MVP

AREA	IN THE MVP	PLANNED NEXT
Notifications	In-app, with delivery log	Push, SMS and WhatsApp gateways
Driver app offline	Needs a connection; late GPS pings are accepted	Local queue for the offline stretch
Maps	Schematic stop progress; one-tap navigation	Plant road map with live position
Language	English	Hindi
Integrations	HRMS, SSO and SAP simulated (seeded employees, password login, CSV export)	SSO/AD login, HRMS sync, SAP posting
Booking time	Five quick choices in the app	Free date and time picker
Optimisation	Rules and scoring on configured leg times	Learned ETAs, demand forecasting, route solver

Scale and operations	Single MySQL database; demo data reseeded on start	Schema migrations, GPS store and cache, load test at shift change
Security hardening	Suitable for a demo	Production secrets, restricted origins, httpOnly web session, refresh tokens, demo reset off

How the scope was verified

CHECK	RESULT
Backend tests: unit, web slice and MySQL integration, including a race for the last seat	`43 / 43 pass`
End-to-end API check of the full trip	`43 checks pass`
Scripted walk of every screen in this document on the web build	All screens reached; screenshots in this document come from that run
Recorded journey videos (user, driver, admin)	All three complete without errors

Open item found in review. On Routes & rules, the green "Saved" confirmation disappears immediately after a successful save. The change is saved and audited; only the confirmation is lost. Small fix, scheduled with the next change.